

ARINDAM SINGHA

GenAI / LLM Engineer • Agentic AI & RAG Systems • LangChain •
LangGraph • MCP

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GitHub | Open to roles across India

PROFESSIONAL SUMMARY

Generative AI and Machine Learning Engineer with approximately 10 years of combined industry and research experience, including 4+ years in automotive AI and a Ph.D. in intelligent robotic systems. Builds production-oriented LLM applications, RAG pipelines, LangGraph agents, multimodal document workflows, and MCP-based tool integrations. Delivered an agentic RAG solution that reduced manual test-design effort by 60%. Strong background in Python, deep learning, computer vision, ROS2, cloud deployment, and scalable AI services.

CORE SKILLS

Generative AI	LLMs, RAG, Agentic RAG, AI Agents, Prompt Engineering, Structured Outputs, Fine-Tuning, PEFT/LoRA
Frameworks	LangChain, LangGraph, MCP, Hugging Face, OpenAI API, Ollama, Gradio, Streamlit
Retrieval	Vector databases, embeddings, semantic search, chunking, reranking, multimodal RAG, ChromaDB, FAISS, Pinecone
ML / Engineering	Python, PyTorch, Transformers, YOLOv8, computer vision, FastAPI, Docker, Azure, Triton, gRPC, CI/CD
Robotics	ROS2, Gazebo, Webots, MoveIt2, Nav2, digital twins, sensor fusion

PROFESSIONAL EXPERIENCE

Lead Engineer - AI & Generative Systems <i>Magna Steyr India, Pune</i>	Jul 2024 - Present
- Architected an agentic RAG solution with LangChain and LangGraph to generate software test cases from engineering requirements in CSV, PDF, and Excel formats. - Built multimodal ingestion for engineering documents and infotainment UI images, including chunking, embeddings, semantic retrieval, and iterative chatbot-based refinement. - Reduced manual test-design effort by 60% through an end-to-end LLM workflow and interactive human-in-the-loop review. - Integrated Model Context Protocol (MCP) to standardize tool calling between LLM agents and external services. - Deployed YOLOv8 vision models with NVIDIA Triton in a microservice architecture and helped migrate the platform to ROS2 for LLM-to-robot integration.	
Senior Engineer - Perception & AI <i>Magna Steyr India, Pune</i>	Feb 2022 - Jun 2024
- Developed a Webots and ROS2 digital twin to validate AI perception models before production deployment. - Built modular gRPC microservices for scalable AI pipelines and managed reliable Docker-based deployments on Microsoft Azure.	
Senior Research Fellow <i>National Institute of Technology Sikkim</i>	Jan 2018 - Dec 2021
Applied neural networks to multi-UAV trajectory tracking and formation control under dynamic conditions; published peer-reviewed research on heterogeneous UAV-UGV systems.	
Junior Research Fellow <i>National Institute of Technology Sikkim</i>	Jan 2016 - Dec 2017
Developed grid-based navigation methods for autonomous mobile-robot exploration and mapping.	

SELECTED GENERATIVE AI & MACHINE LEARNING PROJECTS

Requirement-to-Test-Case Generation LLM, RAG, LangGraph	2025 - Present
End-to-end agentic application for engineering test generation, combining CSV/PDF/Excel ingestion, image understanding, retrieval, agent orchestration, and conversational refinement.	
MCP Weather Service MCP, Gradio	2025
Implemented an MCP server and interactive UI to demonstrate reliable LLM tool calling, execution, and agent-workflow visualization.	
Industrial Defect Detection Platform YOLOv8, Triton, Docker	2022 - 2024
Contributed to a production vision platform for automated manufacturing inspection, including model deployment, microservices, and performance-focused inference.	
Real-Time PPE Detection YOLOv8, Streamlit	2024
Built a real-time industrial safety application with video inference, visualization, and alert management.	
Traffic Signal Detection Computer Vision	2024
Developed a real-time traffic-light state detection pipeline for ADAS and intelligent transportation decision support.	

EDUCATION

Ph.D., Intelligent Networked Robotic Systems National Institute of Technology Sikkim	2016 - 2021
M.E., Electronics, Instrumentation & Control Thapar University	2012 - 2014
B.Tech., Applied Electronics & Instrumentation Engineering Abacus Institute of Engineering and Management (MAKAUT)	2008 - 2012

SELECTED PUBLICATIONS

- A. Singha et al., "Adaptive Neural Network based Quadrotor UAV Formation Control under External Disturbances," *Aerospace Science and Technology*, 2021.
- A. Singha et al., "Self-Organizing Map based Strategic Placement and Task Assignment for a Multi-agent System," *Advances in Intelligent Systems and Computing*, 2021.
- A. Singha et al., "UGV-UAV Coordination: Take-off, Landing and Formation Control," IEEE World Automation Congress, 2021.
- Author of 13 peer-reviewed publications; complete list available on ResearchGate.

AWARDS & RECOGNITION

- Visvesvaraya PhD Scholarship - Digital India Corporation, Ministry of Electronics & IT, Government of India
- MHRD GATE Scholarship - Ministry of Human Resource Development, Government of India
- Team of the Month - Magna Steyr India (Oct 2023, Apr 2025); Employee of the Month (Dec 2022)

ADDITIONAL INFORMATION

Languages: English (professional), Hindi (fluent), Bengali (native)

Work location: Open to GenAI, LLM, Applied AI, and ML engineering roles across India; open to hybrid, on-site, or remote arrangements.